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(1) Research Program/Project/Study Title: Education in the midst of Pandemic: TED Students' Readiness and Challenges in Modular Learning	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☒ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation			
		Municipality/City:	Dumarao
		Province:	Capiz
		Region:	VI
(5) Classification		(6) Mode of Implementation	
Research:	Development:	☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
☐ Basic	☐ Pilot Testing		
☐ Applied	☐ Tech. Promotion/Commercialization		
(7) Sector/Commodity		(8) Discipline/Field of Specialization	

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(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

Face-to-face learning between students and instructors has been halted throughout the institution. There is a move toward the modular approach in teaching that has drawn a special attention in the country not only in the basic education unit but also in the higher education. The key purpose of this study was to find out the readiness and challenges of the students towards modular learning approach. It seeks to answer the following questions; What is the socio-demographic profile of the respondents? What is the level of students' readiness and challenges in using the learning modules? Is there a significant relationship between the students' readiness and challenges? The study utilized descriptive quantitative research design, the data were analyzed using the descriptive statistics such as frequency count, percentage, mean and inferential statistics such as standard deviation, Pearson R in determining the relationship between the readiness and challenges of modular learning in the midst of pandemic. The data revealed that respondents were "ready" ($M=3.16$, $SD=0.80$) to take the modular learning approach in terms of mental, physical and emotional aspect. On the otherhand despite of being "ready" they were challenged upon learning on their own with a mean score of 3.66 considering some factors namely; internet problems, financial capacity and knowledge on technology. It was found out that there was positive very weak linear relationship and significant ($r=0.385$, $p=0.001$). The students must continue to be intrinsically motivated to face the changes in the learning approaches, they must continue to be technologically equipped and used their gadget to acquire knowledge in this time of pandemic.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

Readiness, Challenges, Modular Learning Approach

(10) Introduction

Teaching is a human activity. There is an interaction between the teacher and the learner. It is a skillful activity, and applies knowledge, and scientific principles to the purpose of setting an environment to facilitate learning. There are some modifications that took place in the educational system not only in the Philippines but in some countries.

Now adays, students and teachers must be ready in facing the new changes in the educational process, especially during this time of COVID 19-pandemic that the world are experiencing right now. Face-to-face learning between students and instructors has been halted throughout the institution. There is a move toward the modular approach in teaching that has drawn a special attention in the country not only in the basic education unit but also in the higher education. Modular approach is an emerging trend educational thinking that shifts traditional method of instruction to an outcome based learning paradigm.(Zamir, 2014). The trend of using modules as teaching-learning approach is becoming very popular in recent times. The teacher uses teaching modules prepared for specific purposes instead of traditional textbook. Students must learn independently with the guidance of the teacher through text messages, group chats and any form of communication. The researchers find it interesting to study this approach to check the readiness and challenges of the students, for the teachers to improve teaching learning effectiveness for the betterment of education.

This study focuses on determining the readiness and challenges towards the learning modules in the new normal. It seeks to answer the following questions; What is the socio-demographic profile of the respondents? What is the level of students' readiness and challenges in using the modular learning approach? Is there a significant relationship between the students' readiness and challenges?

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(11) Materials and Methods or Methodology

MATERIALS

A set of three-part survey questionnaire was used in obtaining the data. The research instrument had been adopted by the researcher from the study of Garcia, G. 2020 entitled “*Modular Approach in Science: Students’ Readiness and Challenges in the New Normal*” . Part I contain the socio-demographic profile of the respondents. Part II were questions measuring the level of readiness Part III consists of the challenges encountered by the participants.

METHODOLOGY

For this study, the researcher used descriptive quantitative research design to describe students’ readiness and challenges in modular learning approach. This study was conducted during the first semester of the academic year 2021-2022. The respondents of the study were the 68 Teacher Education Department students of CAPSU-Dumarao specifically the second year BEED, BSED Social Studies and General Science. The study adapts the formula used by Slovin in identifying the sample size where:

$$n = \frac{N}{1 + N (e)^2}$$

Where:

n – is the sample size

N – is the population size

e – is the margin of error

In the study, the second year Teacher Education students were determined as one group and categorized according to their course and major, BEEd, BSEd-Social Studies, BSEd General Science. Simple random design was used in determining the respondents. The 68 total of the respondents were divided into 18 from General Science, 35 Social Studies and 15 from the BEEd. After the classification of the course/majors, respondents were chosen randomly. Stratified Random Sampling technique was used in selecting the respondents. This is done by separating the population units into non-overlapping groups determined by every strand and subsamples are selected through simple random sampling. In this study, there were total of 68 numbers of respondents distributed into different course/majors.

The data were analyzed using the descriptive statistics such as frequency count, percentage, mean and inferential statistics such as standard deviation, Pearson R in determining the relationship between the readiness and challenges of modular learning in the midst of pandemic. Considering the Covid-19 pandemic the data were collected to the 68 respondents on Google form.

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(12) Results and Discussions

Table 1. Socio-demographic profile of the respondents

Socio-Demographic Profiles	Frequency	Percentage % 100
Course		
BSEd General Science	18	26.5
BSEd Social Studies	35	51.45
BEEd	15	22.05
Total:	68	100.0
Gender		
Male	14	20.6
Female	54	79.4
Total	68	100.0
Gadgets		
Cp_only	40	58.8
Tv_laptop	1	1.5
Cp_laptop	5	7.4
Tv_cp	16	23.5
Tv_cp_laptop	6	8.8
Total	68	100
Internet Connectivity		
Own_mobile_data	48	70.6
Own_broadband	3	4.4
Other_places	3	4.4
Own_data_broadband	3	4.4
Own_data_other_places	8	11.8
Own_broadband_other_places	3	4.4
Socio-Economic Status		
Low Income	61	89.7
Lower Middle Income	7	10.3

Course- Among the 68 respondents, 18 or 26.5% were came from the BSED General Science, 35 or 51.45 % from Social Studies and 15 or 22.05 % are from the BEEd. Mostly of the respondents were Social Studies since they were more in number.

Gender- Majority of the respondents were female (54 or 79%) while male were (14 or 20.6%).

Gadgets- In terms of gadgets, 40 out of 68 respondents used cellphone only which accumulated 58.8 % as a mode of communication to the course facilitator and also it helps to better understand the learning modules. The result also shows that 1 or 1.5 % used the television and laptop, 5 or 7.4% used both cellphone and laptop ,16 or 23.5% used both television and cellphone while 6 or 8.8 % were utilizing the three gadgets cellphone, laptop and television.

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Internet Connectivity-since most of the respondents have cellphone as a gadget 48 or 70.6 percent had a mobile data in their places, while 8 or 11.8 percent acquired mobile data in other places which means their internet connectivity at home was poor, 3 or 4.4% have owned broadband. Lastly 3 or 4.4% connected their gadgets using the broadband of other places while communicating with their course facilitator.

Socio-economic status- Among the 5 options included in the instrument of socio economic status only the low income and lower middle income came out, which results to 61 or 89.7% and 7 or 10.3% respectively.

Table 2 shows the level of readiness of the respondents to the modular learning approach

Variables	Mean	Description	SD
Mental Aspect	3.26	Ready	0.80
Physical Aspect	3.02	Ready	0.79
Emotional Aspect	3.21	Ready	0.78
Overall Behavior	3.16	Ready	0.80

Scale	Description
4.21-5.00	Very Much Ready
3.41-4.20	Much Ready
2.61-3.40	Ready
1.81-2.60	Quite Ready
1.00-1.80	Not Ready

Table 2 shows the level of readiness and challenges of the respondents to the modular learning approach. The data revealed that respondents were “ready” (M=3.16, SD=0.80) to take the modular learning approach in terms of mental, physical and emotional aspect. This implies that students are open to self directed learning and always consider the current status of the educational system, they are prepared to accept the changes in the teaching and learning process. The availability of the gadgets helps them to be open and be ready for the shifting of face-to-face instruction to modular learning approach. The result conform to the study conducted by Arooj, M. et.al (2021), in spite of curriculum change from traditional to integrated modular curriculum, students showed positive response towards desire for learning which incorporates positive attitude toward studies. However, it contest to the study of Guansi, M. 2020 , states that college students are not prepared in terms of online learning, blended learning, face-to-face learning, and modular/distant learning due to factors. These factors include financial capacity, resources, and knowledge of technology. The readiness of the students contributes to the smooth implementation of the modular learning approach by overcoming the challenges brought by the abrupt change in the teaching learning paradigm.

Table 3 shows the level of challengess of the respondents to the modular learning approach

Variables	Mean	Description	SD
Challenges	3.66	Challenged	0.91

Scale	Description
4.21-5.00	Very Challenged
3.41-4.20	Challenged
2.61-3.40	Neither Challenged
1.81-2.60	Less Challenged

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1.00-1.80 Not Challenged

The table 3 shows the level of challenges of the respondents to the modular learning approach, the data revealed that the students were “challenged” ($m=3.66$, $SD\ 0.91$). Despite that students were physically, mentally and emotionally ready, they were challenged upon learning on their own. The data was supported by the study conducted by Sumaoang and Dangle 2020 states that most of the learners are having difficulty in this new learning modality, 90% of the participants in their study had a hard time answering their modules. The challenges encountered by the students in complying their tasks and understanding the lessons in their modules was also brought by some factors like having internet problems. It was strengthened by the study of Bustillo E. & Aguilos M. (2022) results showed that most students were constrained by many challenges and struggles in complying with the tasks, these include internet connectivity problems, inadequate learning resources, difficulty understanding the module contents and assessment instructions, overloaded remote learning tasks, poor learning environment, and mental health problems. Regardless of difficulties encountered by the students they find ways to cope with it by being positive and be open mindedness.

Table 4 Correlation on the students’ level of readiness and challenges and vice versa

	Mean	Std. Deviation	N	R	Sig(2-tailed)
Readiness	3.16	.75	68	1	0.001
Challenges	3.66	.91	68	0.385	0.001

** . Correlation is significant at the 0.01 level (2-tailed).

The table shows the correlation on the students’ readiness and challenges. It was found out that there was positive very weak linear relationship and significant ($r=0.385$, $p=0.001$). It implies that students were challenged by the different factors like the abrupt change of the learning modality. And yet they were ready to face the changes. Additionally, they were challenged to become ready in terms of mental aspect since they will deal with the self directed learning, hence the skills for accessing reliable source as guide for understanding module is very vital. Students showed positive response towards desire for learning however they were challenged by several factors namely: financial capacity, resources, and knowledge of technology. Thus, students can learn better if they are mentally ready and technologically equipped in modular learning. In terms of emotional aspect, positive emotions of the students like intrinsic motivation greatly help them to cope with the new modality. Moreover, the physical aspect contributed to the readiness of the students to overcome the challenges of the modular learning modality. In addition, according to Lazzaro C, 2012 Physical readiness of students is determined by the skills and ability to carry out the tasks in the process of activity, particularly in education

(13) Conclusions and Recommendations

In regards to the above findings, the following conclusions can be made; firstly, majority of the respondents were female with the frequency of 20.6%, BSEd Social Studies with the frequency of 51.45%, with cellphones and internet connection at their place with a frequency of 58.8 % and 70.6% and has a low income status with a frequency of 89.7%. secondly, it has been found out that students’ were ready to take the learning modular approach with a mean score of 3.16 in terms of mental, physical and emotional aspect. On the otherhand despite of being “ready” they were challenged upon learning on their own with a mean score of 3.66

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considering some factors namely; internet problems, financial capacity and knowledge on technology. Thirdly, It was found out that there was positive very weak linear relationship and significant ($r=0.385$, $p=0.001$). This result infer that the students were ready to face the challenges of modular learning approach.

Recommendations:

With all the conclusions given, the following recommendation is forwarded.

Firstly, The students must continue to be intrinsically motivated to face the changes in the learning approaches, they must continue to be technologically equipped and used their gadget to acquire knowledge in this time of pandemic. Secondly, there is a need to improve the knowledge on technology and internet connectivity of the students in their area. Thirdly, the University may consider conducting more comprehensive researches related to the effects of modular learning approach to the academic performance of the students for the continuity of the study.

(14) Output

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments

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Aizel C. Coroz
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